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Home rule and environmentalism: the adoption of green initiatives in US municipalities

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This article examines the adoption of green initiatives among US municipalities. The author posits that home rule municipalities are more likely to embrace green initiatives than their non-home rule counterparts. Through analysis of an original data set of 50 municipalities, the author finds that the adoption of a home rule charter increases the likelihood of green initiative adoption within a municipality. In conclusion, this research proposes that the consideration of home rule status as an accurate environmental policy predictor needs further scholarly attention and may be a missing link regarding environmental policy adoption at the local government level.

Keywords: home rule; environmental policy; urban policy; local government autonomy; green initiatives

1. Introduction

In recent years, there has been a movement in academia from studying environmental policy on the national level to a much narrower examination of local municipalities and their respective green initiatives. The implications of this research reach well beyond the borders of academics and extend to municipalities and their corresponding local policy-makers. Scholars must ask what particular characteristics of municipalities play the most significant role in green initiative adoption. Recent urban scholarship has suggested several variables that influence the adoption of green initiatives, but one specific variable, municipality home rule status, has had little to no scholarly consideration.

First, defining what constitutes a green initiative is not always straightforward. In some cases, the expression green initiative could be interchanged with other terms such as sustainability or going green. Moreover, these initiatives can be developed and managed by several different actors

ranging from the individual level all the way up to the federal government level. For the purposes of this research though, green initiatives are defined merely as projects or policies managed by a municipality that have the central goal of curbing the human impact on the environment.

Some examples of green initiatives that the municipalities in the research have adopted include (1) recycling and waste programmes that can store and reutilise water run-off or turn trash into compost for organic farming, (2) green building techniques such as harnessing the sun's rays for heating, or constructing green roofs that are completely or partially covered with vegetation, and (3) the development of municipality green space such as park and recreation sites or new community gardens. Although rather basic, this wide-ranging definition covers a large variety of green initiative possibilities and makes research on municipality green initiative adoption less complicated.

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This research posits that home rule municipalities are more likely to embrace green initiatives than their non-home rule counterparts. Through the design and analysis of an original data set, this research will contribute to urban scholarship through the exploration of a relatively new variable in comparison to more commonly used variables in environmental policy analysis at the local level. The essential underpinnings of local environmental policy adoption have already been laid out by previous scholars (e.g. Kahn 2006; Portney & Berry 2010). However, this fundamentally unique research presents a tangible contribution to urban studies by creating an opportunity for further research, focusing on how home rule status impacts green policy adoption.

Over the course of this article, I first examine the extant research relating to home rule and environmental policy on the local level. Following this review, there is a brief discussion of both the methodological approach and the data collection that was applied to test my specific hypothesis regarding municipality environmentalism. Next, there is a description of both the quantitative and qualitative analysis, followed by the inferences that can be drawn from the said analysis. The research concludes with an overview of the findings, their indications relative to current scholarship, and a brief description of the direction in which future research should go.

2. Home rule

Municipal home rule is rather neglected in social science research, most notably because of the numerous variations that exist across the United States as well as the confusion that it can cause to even the most elite scholars. This study attempts to plunge deeper into home rule adoption by analysing further its effect on local government autonomy and policy adoption, which to this point cannot be assessed fully. At the outset, the definition of home rule in the United States, at a rudimentary level, is local government autonomy bestowed upon local governments based on particular conditions that vary from state to state.

Richardson's (2011, p. 663) definition fits this mould precisely as he describes home rule as 'actual grants of authority given to local governments ... found in local and state statutes, both initiated by the state legislature'. From this, home rule can be further described as authority given to local governments to administer their own affairs without state intrusion (Timmons *et al.* 1993). Nearly every US state has a home rule option of varying degree with only six states, Alabama, Idaho, Indiana, Mississippi, Vermont, and Virginia, not providing any form of municipal home rule (CELDF 2013). With home rule being such a commonly used tool for local governments to increase their autonomy, understanding if its adoption may also increase the adoption of green initiatives is worth studying further.

2.1. Home rule and local government autonomy

This autonomy given to cities through home rule adoption usually comes in a combination of these four components: (1) structural authority over what form of government to implement, (2) functional authority or the power to exercise local self-government either broadly or limited depending upon the state, (3) fiscal authority over revenue sources and the regulation of local tax rates, and (4) personnel authority over employment rules, conditions, and collective bargaining (Richardson 2011; NLC 2013). Although each of these four components is widely used among home rule municipalities, the functional powers are the most commonly used, which bodes well for the prospects of home rule having a significant impact on environmental policy adoption (NLC 2013).

As mentioned previously, the autonomy that home rule provides varies depending upon the state in question, and most scholarly attempts at ascertaining a quantitative measurement of local government autonomy relative to municipal home rule has yielded only minor successes (Weeks & Hardy 1984; Krane *et al.* 2001; Richardson *et al.* 2003; Geon & Turnbull 2004; Bluestein 2006). So any attempt to draw direct connections between

local government autonomy and home rule adoption is problematic, however, as it pertains to this research, a brief review of how home rule is utilised may be useful in attempting to connect the two relative to green initiative adoption.

The municipalities in this study derive from five states: California, Illinois, Michigan, Pennsylvania, and Texas. Each of these states has three central home rule related powers in common, providing a generally equal foundation for this research: (1) home rule can be both enacted and rejected through voter referendum, (2) home rule municipalities are able to create and develop their own form of government, and (3) each home rule municipality has the ultimate legislative authority over their municipal affairs. These broad authorities can be classified as both structural and functional powers, as described above, and are critically important to the municipality's autonomy.

One specific example of the authority of home rule units in the state of Texas is the power of annexation (TML 2013). The ability to annex nearby lands creates an invaluable opportunity for home rule municipalities to increase the overall land area of the city as well as use that extension of land for new business centres, residential areas, or even new local parks and recreation areas. On another note, in the state of California, home rule units have the functional power to pass local ordinances, and even more specifically to enact building code ordinances within the municipality (LCC 2013). The prospect of enacting municipal building codes opens up the opportunity to apply green building techniques to all new commercial and residential construction. These are only two of numerous examples describing how home rule can increase local government autonomy and environmental policy adoption, and hopefully this research will indicate whether these opportunities are being utilised in home rule municipalities.

Furthermore, it must be noted that home rule may not be necessary in all cases of municipality green initiative adoption. Some green initiatives, such as several different types of recycling

programmes, can be enacted and implemented without a home rule charter. Whether home rule is needed or not really depends on the type of environmental policy the municipality wants to achieve, and in several instances home rule adoption may not be needed to realise its goals.

3. Independent variable selection

As noted previously, there is strong movement towards research addressing environmental policy and sustainability on the local municipal level. Variable choice among urban scholars represents the widest discrepancy in current green initiative research. One of the many variables considered by urban scholars is the wealth and resources of a municipality that in turn can be applied towards green initiatives. According to many scholars, higher levels of environmentalism within municipalities correlate significantly with the wealth and overall resource availability of a municipality (Press & Balch 2002; Kahn 2006; Lubell, Feiock, and Handy 2009). Through Lubell, Feiock and Handy (2009) creation and examination of their sustainability index, they discovered that although a municipality's wealth and available resources are key contributors to eco-friendly policy, multiple variables have a significant impact including population, socioeconomic status, among others.

A second predicting variable that has been studied extensively is the style of government through which the city is structured. Lubell, Feiock, and Ramirez de la Cruz (2009) proposed that land use regulations vary depending on the power structure of mayors, city managers, and local interests. On a similar note, Jepson (2004) found that community planners and developers have a substantial impact on local level environmentalism. Each of the above authors found that the local governmental structure plays a critical role in environmental regulation and policy, albeit through different facets of sustainability and governance. Furthermore, the type of government structure a municipality employs is likely to impact not only what types of green initiatives

are adopted, but also the amount of green initiatives that are adopted. For example, will the more professionalised council/manager form of government be more likely to adopt green initiatives? Or could it be the more localised mayor/council format that provides a higher possibility for green initiative adoption? This research will attempt to develop some preliminary associations between the style of local government and green initiative adoption.

Multi-level governance has also been found to have a significant impact on environmental initiatives on the local level according to some scholars (Betsill & Bulkeley 2006; Rabe 2008; Betsill & Rabe 2009). Betsill and Bulkeley (2006) performed an analysis of the multi-level governance body entitled Cities for Climate Protection. The authors used evidence from this case study to explain the variance among local environmental initiatives, and argued that multi-level governance explains local environmental policy adoption better than any other variable or model. The authors argued that dividing up the levels of governance prior to research both limits and predisposes the potential outcomes. For this reason, the authors believe that an examination of both local and state governance should be considered in order to achieve an accurate and complete understanding of environmental policy adoption.

Increased civic participation has also been shown to have a positive relationship with environmentally sustainable policies on the municipal level (Portney 2005; Portney & Berry 2010; Krause 2012). Portney and Berry (2010) specifically found that those cities that take sustainability more seriously have a positive correlation between high levels of participation and pursuit of environmentally sustainable policies. Theoretically, the more civically knowledgeable local citizenry is of their government and environment, the more likely the municipality will address environmental issues with green initiatives. However, the authors did acknowledge that civic participation alone does not explain the wide variance in local environmental policy adoption. This suggests that urban scholarship must continue to study and search for

original variables that are relative to environmental, sustainable, or green policy adoption.

A municipality's mean education attainment has also been shown to correlate with the adoption of green initiatives (O'Connell 2008; Portney 2008). O'Connell (2008) used ordinary least squares (OLS) regression of survey data to show that a city's smart growth policy is significantly linked to their corresponding percentage of college level graduates. This finding may indicate that the knowledge attained from higher education directly influences feelings towards environmental policy adoption. So, as O'Connell (2008, p. 1638) suggests, 'cities with more college graduates may contain more people who support smart growth'.

One variable in particular has obviously been overlooked within recent scholarly literature as it pertains to green policy. Home rule is disregarded almost wholly from scholarly discussion, which in turn prompted me to further examine the influence that it may have on green initiative adoption. The municipality's status of home rule or non-home rule, as Richardson (2011) suggests, does have at least a minor effect on local government autonomy, and therefore would also affect the possibilities of and opportunities for green initiative adoption and implementation.

4. Methodology and data

First, the hypothesis that home rule municipalities will have higher levels of environmentalism is addressed through quantitative analysis. The supposition is tested using binary logistic regression to explain the probability of this premise being due to random error. These quantitative methods will provide conclusions that are empirical in nature, providing more efficient findings, which can in turn be functional for both academic scholarship and local government policy analysis. However, quantitative analysis can misrepresent data by not accounting for contextual information that other methodologies could, and because of this a brief qualitative section including case studies of two chosen municipalities from the developed data set will provide that missing contextual link to the

logistic regression analysis. These methods are only as strong as the data selection however, and since there is no prevailing uniform data set for this particular research, a new data set was developed based on the evaluation of city and state governmental websites. The accuracy of this data set is reliant upon municipality self-reporting which many scholars may deem unpredictable, but even so, this data set is not meant to be a perfect creation. This research attempts to only provide a new perspective to be applied towards the adoption of green initiatives not yet represented in urban scholarship.

The data set itself examines 50 municipalities deriving from five states: California, Michigan, Illinois, Pennsylvania, and Texas. The states were first chosen to represent each region of the United States, North, South, East, West, and Midwest with the specific purpose of eliminating regional predispositions towards green policy adoption and second because they all are relatively large populous states. As mentioned in an earlier section, each of these states share three important characteristics regarding home rule, but one that sets these states apart from several others is the ability to both enact and reject home rule charters through voter referendum. Each municipality within the five states was chosen at random, with the only control being the intentional collection of both 25 home rule and 25 non-home rule municipalities in order to provide an equal state upon which to conduct analysis. To ensure municipality randomisation two lists were created for each of the five

states, one composed of all the home rule municipalities and the other with all the non-home rule municipalities. Then 25 municipalities from each of the 10 lists were chosen indiscriminately for inclusion in the data set. Although the concept of home rule does exist elsewhere, this study focuses solely on municipalities in the United States. Home rule in Europe and other parts of the world tend not to have the same legal and political issues with home rule charters that the United States does.

On another note, the variables indicated and developed include: educational attainment, defined as the percentage of the population with a bachelor's degree or higher, population, mean income, form of government defined as the most closely linked government structure to the following: (1) mayor/council and (2) council/manager, home rule status, measured as a dummy variable, and the environmental policy adoption scale. A descriptive summary of the independent variables can be seen in Table 1.

Furthermore, the environmental policy adoption scale was determined through the combination of three separate variables that originate from the Smarter Cities Project conducted by the Natural Resources Defense Council. The variables include (1) whether or not the municipality offers a recycling programme, (2) the amount of green space in acres, and (3) whether or not green building techniques are commonly used within the municipality. Green space was demarcated as park and recreation total acreage, and green

Table 1. Descriptive statistics.

Variable	Description	Mean	Std. Dev.	Min.	Max.
Population	Total population of the municipality	3.99×10^4	4.19×10^4	1555	155,937
Home rule status	Non-home rule (0) or home rule (1)	0.50	0.505	0	1
Form of government	Mayor/council (1) or council/manager (2)	1.58	0.49	1	2
Education attainment	Percentage of population with bachelor's degree or higher	26.65	14.86	4.9	79.4
Mean income	The mean income of a household in the municipality	6.81×10^4	2.47×10^4	32,931	170,096

building techniques were defined using the standards provided by the Leadership in Energy and Environmental Design (LEED). As for recycling, the municipalities in this analysis have introduced several different forms, kerbside recycling pickup, local recycling drop-off centres, community recycling containers specifically for aluminium or plastic, and waste composting to name a few. However, this research does not differentiate between the different types of recycling programmes, only whether the municipality offers one or not. Moreover, these three policy areas were chosen because they indicate the early foundations of a municipality moving towards environmentalism. Even though the policy areas chosen here do not directly isolate those municipalities with the most green initiative implementation, they do signal which municipalities are at least in the beginning stages of adopting environmental initiatives.

The combination of these three variables ultimately equated a score from 0 to 4, with 4 being the highest amount of environmentalism and 0 being the lowest. Table 2 indicates the precise measurement used for each of these three variables. For the purposes of this study's implementation of logistic regression, the environmentalism scale of 0 to 4 was reduced into two categories. Those municipalities that scored 2 or below on the original environmentalism score were coded as one for low environmentalism, and those scoring 3 and 4 were coded as two for high environmentalism. This was done not only to create a two-category dependent variable for accurate logistic

regression, but also because almost all municipalities have adopted some form of a recycling programme. Therefore, there is no useful basis for including the recycling variable in the final environmentalism assessment.

This data set provides a new combination of determinants of environmentalism within a municipality, but of course these determinants can be criticised because there are many other indicators of an eco-friendly city. Including every variable would create a data set too complex, possibly resulting in a loss of parsimony that keeps the findings easy to comprehend for practical purposes. This data set also provides an *N* of 50, eliminating partially the bias of a single case study. Most urban scholarship studies highly populated and prestigious cities, but this data set, on the contrary, includes municipalities of all populations and sizes further extending the scope of urban studies to smaller normally ignored towns and villages. Figure 1 displays all 50 municipalities and their corresponding population sizes further exhibiting the wide-ranging populations. In sum, this methodology and data should lead to findings that will further urban scholarship and the understanding of local environmental policy adoption.

5. Correlation and logit analysis

To begin, a basic correlation matrix is displayed in Table 3. Of the five independent variables included in this analysis, the mean income variable displays the most significant correlative

Table 2. Environmental policy adoption scale measurement.

Municipality recycling		Municipality green space (in acres)		Municipality green building (LEED)	
Yes (1)	No (0)	1	2	Yes (1)	No (0)
Municipality offers city wide recycling programme	Municipality does not offer city wide recycling programme	0–999	1000 and above	Municipality reported usage of green construction techniques	Municipality did not report usage of green construction techniques

Note: Measurements based upon municipality self-reporting on their official website.

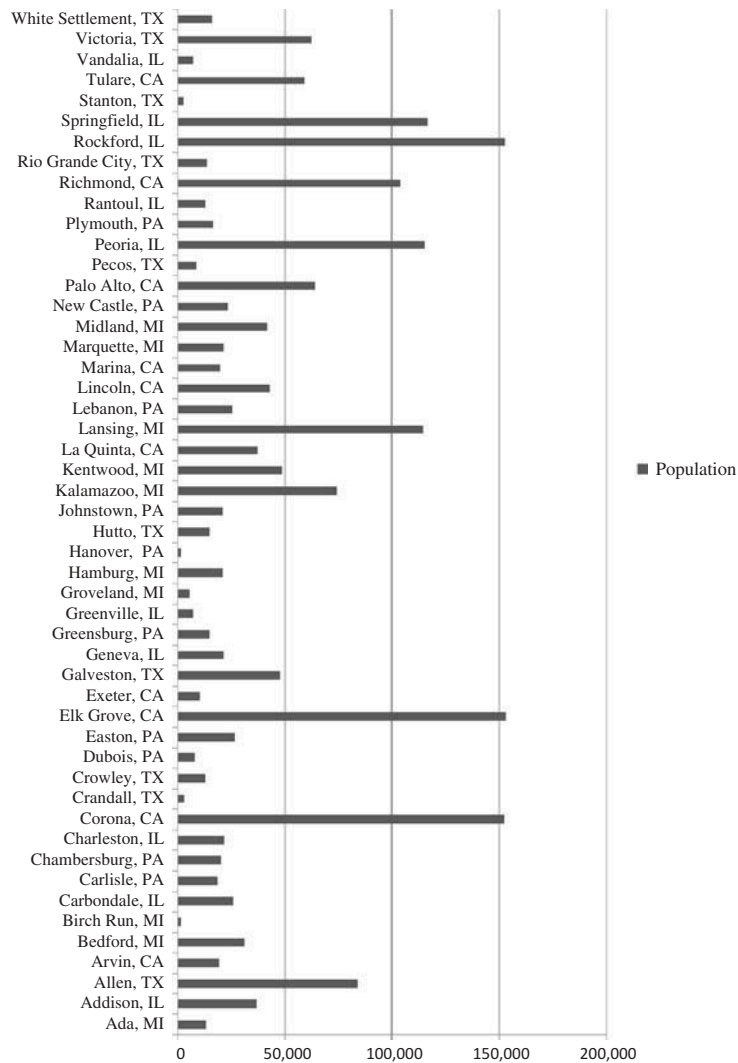


Figure 1. Municipalities by population.

Table 3. Correlation matrix.

Variables	(1)	(2)	(3)	(4)	(5)
(1) Population	1				
(2) Home rule status	.149	1			
(3) Form of government	.175	.203	1		
(4) Education attainment	.189	.131	.224	1	
(5) Mean income	.076***	.007	.312**	.761*	1

Note: * $p \leq .01$; ** $p \leq .05$; *** $p \leq .1$.

relationships. More specifically, mean income demonstrates a positive, statistically significant relationship with three other predictor variables, population, form of government, and education attainment. In other words, the higher the population and the higher the educational attainment of a municipality, the higher the mean income will be. However, both of these relationships are rather weak correlations.

The third and final statistically significant correlation is not only the strongest, with a Pearson correlation coefficient of .761, but is also the most thought provoking. As the mean income of a municipality increases, so does the likelihood that the municipality has a council/manager form of government. This basic finding may indicate that the council/manager form of government, most closely associated with a professionally trained city manager who runs the city similarly to the chief executive officer (ICMA 2014), may be managing the city with more financial awareness than that of a mayor who holds less professional training. A correlation matrix alone cannot accurately depict this relationship alone however, more analysis of this preliminary finding is necessary in future research.

Moreover, the logit analysis findings are represented through four regression tables displaying the outcome of the original hypothesis, as well as the other conditional variables selected from the data set. The regression equation used multiple independent variables evaluating each variable's statistical significance relative to the dependent variable. The pseudo- R^2 value, as the note under Table 4 shows, estimates the amount of variance

explained by the regression equation, and is equal to .592. Therefore, we can estimate that this regression equation explains about 59% of the variance in municipality adoption of green initiatives. This however does leave open a window for other variables to have significance such as, multi-level governance, community social capital, civic participation, among others. This gap, however, is not a wide one, and ultimately this regression equation explains a greater part of the variance. Table 4 also displays this particular model's high χ^2 value and corresponding significance level allowing for the rejection of the null hypothesis, and demonstrating that this regression model fits the observed data quite well. A third measure, the proportional reduction of error (PRE), calculates to a 52% reduction in error when predicting the dependent variable based on the inclusion of the independent variables used in this logistic regression equation. These three critical statistical measures indicate the high level of accuracy and worth of this regression equation in explaining at least some variance associated with green initiative adoption on the municipal level.

Table 4 also represents each of the independent variables that were included in the logistic regression equation and signifies their correlative significance in relationship to each municipality. The statistical significance of each predictor shows the impacts that they have on the environmental policy adoption scale. Through this regression analysis, we can see that three of the five predicting variables do not play significant role in explaining any of the variance. The mean income, education attainment, and form of government

Table 4. Logit analysis of determinants of municipality environmentalism.

Independent variables	B Term	S.E.	Sig.
Population*	2.335	0.721	.001
Home rule status**	1.845	0.848	.029
Form of government	1.043	0.917	.255
Education attainment	0.286	0.728	.695
Mean income	-0.626	0.828	.450

Notes: The dependent variable is coded '1' for low environmentalism and '2' for high environmentalism. * $p \leq .01$; ** $p \leq .05$. $\chi^2 = .000$. Pseudo- $R^2 = .59$. PRE = .52.

variables do not appear to have a significant impact on green initiative adoption with each of the corresponding significance levels rising well above the needed .05 significance level. These findings do indeed have implications for both academic scholars and local level policy-makers, in that these variables may not need to be valued heavily when considering green initiative adoption. Two of the independent variables, home rule status and population, are both statistically significant with a 95% certainty that their significance is not due to random error.

In sum, the logistic regression analysis supports the original hypothesis. The significance of the home rule status predictor is equal to .029, confirming the municipalities that have enacted home rule are more likely to adopt green initiatives. This is not to say that home rule makes municipalities greener, but that the adoption of a home rule charter helps eliminate green initiative impediments brought on by other levels of government, most notably the state government. Home rule only becomes important when a municipality wants to enact some form of green initiative. For example, a municipality may want to adopt a new municipal building code with green building techniques, but is unable to do because of state regulations. Home rule, in this case, may be a practice that can do away with possible state government legal obstructions. Even though home rule status does appear to explain a strong portion of the variance, the population variable may even play a larger role.

The statistical significance found for the predictor population equates to .001, which is substantially larger than the home rule predictor. Green initiative adoption has a positive correlation with the population predictor meaning that higher populated municipalities are shown to be more likely to adopt green initiatives. Accurately explaining this relationship is difficult in the least, but some possible inferences can be conjectured.

First, municipalities with larger populations are much more likely to adopt home rule. Many, if not most, US states that have a home rule

charter option allow municipalities to automatically become home rule municipalities once they reach a certain population. From this perspective, population may be acting somewhat as an antecedent variable in its relationship with green initiative adoption. Second, as shown on the correlation matrix in Table 3, the mean income variable tends to rise when the population of a municipality rises. Although this is a weak correlation and only significant at the .1 level, several other scholars (e.g. Press & Balch 2002; Kahn 2006; Lubell, Feiock, and Handy 2009) have found that higher municipality wealth is associated with green initiative adoption. Although the logit analysis in this research does not support this finding, other wealth variables such as the municipality's budget or level of taxation could provide different findings. Furthermore, although my hypothesis was shown to be accurate, municipality population, according to this regression analysis, may have a stronger connection to green policy initiative adoption on the local level.

Slightly different outcomes occurred when the environmental policy adoption scale was separated out into its three original variable forms, recycling, green space, and green building. Each of these variables is represented through three separate logistic regression equations detailed below in Tables 5–7. During this analysis, we find that none of the tested independent variables are significantly related to the adoption of a recycling programme by municipalities. Nearly all of the

Table 5. Logit analysis of determinants of municipality recycling.

Independent variables	B Term	S.E.	Sig.
Population	18.450	4965.468	.997
Home rule status	−0.883	1.789	.621
Form of government	1.826	2.004	.830
Education attainment	18.120	4607.097	.997
Mean income	0.991	1.410	.482

Notes: The dependent variable is coded '0' for no recycling programme and '1' if the municipality has a recycling programme. No significant independent variables. $\chi^2 = .068$. Pseudo- $R^2 = .51$. PRE = 0.

Table 6. Logit analysis of determinants of municipality green space.

Independent variables	B Term	S.E.	Sig.
Population*	4.578	1.853	.014
Home rule status*	3.567	1.630	.029
Form of government**	3.626	1.897	.056
Education attainment	1.728	1.346	.199
Mean income**	-2.681	1.579	.089

Notes: The dependent variable is coded '1' for green space acreage of 0-999 and '2' for green space acreage 1000 and above. * $p \leq .05$; ** $p \leq .1$. $\chi^2 = .000$. Pseudo- $R^2 = .80$. PRE = .60.

Table 7. Logit analysis of determinants of green building techniques.

Independent variables	B Term	S.E.	Sig.
Population*	0.878	0.650	.025
Home rule status*	2.136	0.840	.011
Form of government	0.266	0.820	.727
Education attainment	-0.457	0.645	.131
Mean income	0.873	0.745	.241

Note: The dependent variable is coded '0' for no green building techniques and '1' for green building techniques being utilised. * $p \leq .05$. $\chi^2 = .001$. Pseudo- $R^2 = .49$. PRE = .41.

municipalities in the study, barring only three, have adopted a recycling programme, and therefore, this result further supports an earlier claim that the recycling variable is relatively useless in explaining green initiative adoption overall. Another variable, form of government, was found to be a partially significant variable relative to green space with its significance level reaching .056, just nearly missing the .05 significance level. Continued research and the addition of more cases may prove that the form of government variable is significantly related to the establishment of green space.

The more fascinating and thought-provoking discovery, as it pertains to this research, is found through the analysis of the green building technique variable. Home rule status, as expected, plays a significant role in explaining green building techniques, but interestingly, home rule status

plays a larger role than municipality population in this regression equation. Rather than population explaining the most variance, as shown in the original logit equation, it is home rule status that explains the greatest variance of green building adoption. Home rule status has a positive relationship with this environmental variable meaning that home rule municipalities are more likely to adopt green building techniques than non-home rule municipalities. Moreover, home rule may provide the ability to sidestep possible state regulatory barriers to adopting and implementing green building ordinances that may otherwise make green building construction costly and time-consuming. Only future research can confirm or deny this possible inference, but it is definitely noteworthy. This insight is especially fascinating in that no other independent variable was as significant as home rule status in predicting green building adoption, making this logistic regression equation critical to the home rule variable's possible future inclusion as an important policy predictor.

6. Case study analysis

The following case studies represent two of the municipalities included in the data set, both measured as highly environmental, but with the one distinct difference being home rule status. Examining these two municipalities more closely should support the previously discovered quantitative results by specifically studying the environmental paths of these two different cities, and searching for both their commonalities and differences regarding green initiative adoption.

6.1. Allen, Texas

Allen is located in Collin County, Texas on the north-eastern side of the state about 25 miles north of Dallas and the city's population reached 84,246 in 2010, according to Census Bureau data (2012). Allen was built upon the railway industry which really began to flourish in the late 1800s, and was incorporated as a city officially in 1953 with only

400 to 500 in population. It has since grown dramatically due to the construction of a US highway, an international airport, and being so close in proximity to the constantly developing Dallas, Texas (Allen 2012).

Following the city's declaration as a home rule chartered city in 1979 under the council/manager form of government, there was a population spike that had not been seen at any time prior. From 1980 to 1990, there was about a population increase of 11,000, in part of course due to the surrounding growth of the nation, but also possibly in part due to the home rule charter adoption (US Census 2012). As the population increased, so did local businesses culminating in the retail and local business centre it is today. The dramatic increases in population, local business, and economic growth Allen has today may also be in part due to the 19 straight years of lowered taxes producing quite the incentive for more commercial and private business growth to continue (Allen 2012).

As the city thrived into the 2000s, an enormous growth in environmental initiatives became obvious as the continued development of recreational parks and hiking trails grew to over 800 acres of park land, and 50 miles of hiking trails. The thriving business community has become critical in employing green building technologies, specifically LEED certified projects of many kinds. These green initiatives have been advanced through numerous environmental groups and clubs including the Allen Garden Club, the Keep Allen Beautiful programme, along with many water conservation opportunities for the public (Allen 2013). These group opportunities have encouraged further education of and more public support for green initiatives within the city of Allen. Prior to all of these advancements and green initiatives was the adoption of a city wide recycling programme collecting numerous recyclable goods, and for the items Allen does not collect, nearby locations for recycling drop off are offered (Allen 2012).

At what exact point was Allen on its way to green initiative adoption is hard to pinpoint, and proving even more difficult is what particular

variable was the key contributor to such an increase in local environmentalism. It may have been the exponentially growing wealth that allowed for environmental projects to begin taking shape, and of course the wealth derived from the increase in business and industry due to considerable population growth. It appears Allen is a city that fits the multiple variable effect seamlessly, with everything from population, wealth and resources, and local industry playing a role in the adoption of green initiatives. However, what can be said is that home rule enactment in 1979 opened the door to a population increase, as well as new, less constrictive ways to conduct local economic business, and in turn growing so rapidly. The adoption of a home rule charter, in the case of Allen, Texas, does seem to have played a role over the long run, although an arguably indirect role, in their high levels of green initiative adoption.

6.2. Geneva, Illinois

Geneva is located on the western edge of the Chicago suburbs and as of 2010 had a population of 21,495 (US Census 2012). Geneva has been the county seat of Kane County since its inception in the early 1800s, and therefore has had many prominent county leaders including both lawmakers and businessmen, among others, supervising and encouraging its growth. The city began as a predominantly agricultural village producing such goods as butter and cheese specifically. After officially becoming a city in 1887 and adopting the mayor/council form of government, the railway industry became a key component of local economic growth and business. The city has benefited in more recent times from US highways connecting the city to Chicago, along with the relative closeness of international airports bringing in tourist traffic, and many types of private and commercial business (Geneva 2001).

Important though is the fact that Geneva has not enacted a home rule charter since its incorporation as a city in the late 1800s. Home rule has been discussed and considered throughout the 2000s in city wide meetings, but nothing tangible

has come from those considerations to believe that home rule adoption will occur any time in the near future (Geneva 2012). Even so, Geneva has been growing in terms of environmental initiative enactment, including about 30 city sponsored parks, a recycling programme, and the use of green building techniques. Geneva is measured on the low end of high environmentalism being measured as a three instead of a four on the original environmental policy adoption scale. It may be that critical difference between those cities coded as three and those coded as four is directly related to the enactment of a home rule charter. Home rule status could open up the possibility for further growth in environmental policy and initiative adoption (Geneva 2012).

Similar to Allen, Texas, Geneva's population grew due to its proximity to a larger commercial city, and the nearby railway industry critical for the transportation of both agricultural and manufactured goods. However, Geneva's population is currently only one-third the size of Allen, Texas, and since total population has been found to be a significant predictor of green initiative adoption, this population difference cannot be ignored. It may be that geographical luck played a part in the successfulness of the city, as it indirectly caused an increase in population and wealth, but are those the key contributing factors to Geneva's green initiative adoption? Again, it appears we have a case in which multiple factors played a role in the development and implementation of green initiatives, but Geneva's smaller population and lack of a home rule charter may have created a metaphorical glass ceiling for green policy adoption. If the recent discussion of a home rule charter in Geneva becomes a reality, time will tell whether this city may benefit environmentally from the additional local autonomy.

7. Implications of the findings

With the analysis completed, and the findings presented, it is time to examine the practical inferences that can be drawn from this research in regards to both academic scholarship and

local government policy-makers. As my hypothesis suggested, disregarding the home rule variable, which urban scholars have for the most part done, whether it be intentional or not, may be a major error. Ignoring the home rule variable is discounting an important piece to the environmentalism puzzle, and without it, the mystery of green initiative adoption will never be fully understood. Further research using the home rule variable will change some of the popular environmental policy adoption belief systems within urban studies, and simultaneously alter how urban scholars examine policy adoption in general. More extensive research on the relationship between home rule status and green initiative adoption may continue to show a relationship between the two.

The conditional variable analysis has shown that three particular variables do not show any relative significance to green initiative policy adoption on the municipal level. The mean income, education attainment, and form of government variables may not play as significant of a role as some scholars suggest. Scholars, such as Lubell, Feiock, Ramirez de la Cruz, and Jepson, may need to re-evaluate the impacts that form and structure of government have on green policy adoption. The same goes for those scholars that suggest municipality wealth and municipality education are significant predictors of green initiative adoption. All of these scholars may need to further examine the possible impacts of home rule status along with population in order to get a more complete picture of green initiative adoption.

The predictor, population, was found through this analysis to have a strong significant impact on green initiative adoption. These results illustrate that urban scholarship needs to dig deeper into research, examining why a larger population equates higher levels of green initiatives as this research suggests. It may be that civic participation is more common in larger populated municipalities, or wealth and resources may be more abundant in larger municipalities. No matter which particular variable is the key contributor to green initiative adoption, more expansive research

in these specific areas is the only way to further explain and understand this discovery.

Unfortunately for the local governmental leaders and policy-makers of lower population municipalities, this finding may be damaging to their opportunity for green initiative development. It is not feasible for a small municipality with a small population to increase their population in order to attain green initiatives. Since exponentially increasing population is not a realistic goal of smaller municipalities, the significance of population relative to green initiative adoption may only be a conditional relationship dependent upon other variables such as civic participation, municipality wealth, or the mean education attainment of its population.

As for the case study analysis, we see that in two cities separated by population, geography, and home rule status, successful green initiative adoption has occurred, but to different degrees. Allen, Texas was more successful at green initiative adoption than Geneva, Illinois even though they share so many common historical landmarks. Each city was built upon the railway and transportation industry, resulting in a spike in population, wealth, and resources. The key differences include home rule status, total population, and possibly the environmental programmes and clubs offered.

It appears that home rule status and total population are both significant predictors of green initiative adoption in both of these cities. Following the enactment of home rule, the city of Allen grew tremendously and in turn developed the nationally recognised environmental initiatives they are known for. Allen's population has increased dramatically since home rule charter adoption accumulating to 84,246 in 2010, and as the logit analysis has suggested, the larger the population of a municipality the more likely they are to adopt green initiatives. Geneva has not produced nearly as many green initiatives as Allen, and it appears that their lack of a home rule charter and smaller population, are at least in part, predictors of their lower environmentalism level.

On another note, the city of Allen offers many environmental groups and programmes for the

public to join, while Geneva on the other hand has not. This finding may have some validity, however, because the logit analysis did not include civic participation as an independent variable, further analysis may show that civic participation may significantly affect the presence of municipality green initiative adoption.

A concluding inference that can be drawn exclusive from the output that the case study and regression analysis produced lies within the data set created for the quantitative research. Included in the N portion of the data set were 50 cities ranging from villages to cities. The contribution that this data set analysis provides is an evaluation of overwhelmingly disregarded municipalities that are generally thought of as insignificant to the overall scheme of urban studies. Examining smaller, less populated municipalities provides a new window that prior to this research has, for the most part, been omitted. Continued research on less considered municipalities will create either contributions to current urban scholarship, or original conclusions that apply to the aforementioned smaller municipalities.

8. Conclusion and discussion

In sum, this research has found several new interesting findings relative to local adoption of green initiatives. Of course, this research has not been completed without flaw, and continued research using similar and dissimilar variables, a new definition of environmentalism, and a larger N study could prove to contribute greatly to urban scholarship.

With the addition of variables such as civic participation, the amount of social capital, and possibly the extent of multi-level governance, a clearer and more complete picture of green initiative adoption at the municipal level might be established. Civic participation would be an excellent addition to further research in this area, as it has been found to be a central indicator of a green municipality in previous research. Social capital and multi-level governance have not been examined thoroughly enough up to this point, so the

inclusion of these two variables is also necessary in further policy research. The accumulation of these variables along with the two statistically significant variables in this research, home rule status and municipality population, may provide a greater R^2 value and explain a higher percentage of the variance associated with green policy adoption.

Applying a different definition of environmentalism to this research study could also alter the findings and implications. Adding variables such as air quality or energy production as supplementary variables to the environmentalism calculation would provide a more comprehensive definition of environmentalism as it pertains to municipalities. Green building techniques, measures of green space, and recycling programmes are only initial estimates of a green municipality, and supplementary variables, including the two aforementioned, would measure more wholly environmentalism on the local level.

Another third alteration in the research could be changing the N either in number or in case selection. A larger N study may support the findings of this paper, or provide new significance for a variable or variables that have been overlooked or found to not have an impact in this research. The overall random selection of cases, as for the most part this study implemented, does produce and overall unbiased case selection, but controlling the N for some variables, such as population, would provide a slightly different output of findings.

Lastly, choosing different specific municipalities for the qualitative case study section would provide new and useful insights that the current research does not. Examining home rule municipalities that were not successful in their green initiative endeavours would show under what conditions these municipalities were unsuccessful, and whether certain decisions made by local government negatively affected the possibility of green initiative adoption. If particular business or economic decisions were made differently, would there have been a greater opportunity for green policy? The only way to answer these types of

questions completely is to continue the case study analysis including more municipality variation.

The concluding results show that although home rule status is a significant predictor of green initiative adoption, municipality population plays a somewhat stronger role in green initiative adoption. However, not including home rule in the assessment of a municipality's level of environmentalism may well leave us with an incomplete picture. Urban scholarship can expand upon this research in many above-mentioned ways, but more specifically; this research provides an additional element, upon which improved arguments for municipality green initiative adoption can be formulated.

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